

# SAFETY DATA SHEET



## THYPAAN FERRO

|         |                |             |                                 |
|---------|----------------|-------------|---------------------------------|
| Version | Revision Date: | SDS Number: | Date of last issue: -           |
| 1.1     | 18.12.2025     | 50002035    | Date of first issue: 18.12.2025 |

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### 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product name : THYPAAN FERRO

#### Manufacturer or supplier's details

Company : FMC Agro Kazakhstan LLP

Address : Timiryazev str. 28V, office 501, 5 floor  
050040 Almaty  
Kazakhstan

Emergency telephone number : +44 20 3885 0382 (CHEMTREC's European Regional Toll-Free Number)  
1 703 / 741-5970 (CHEMTREC - International)  
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical Emergency Number : Ambulance in the Republic of Kazakhstan: 103 (24/7, toll-free).  
Almaty City Clinical Hospital, Toxicology Department, 96 Kazybek Bi Street: +7 727 317 01 69 – hotline; +7 727 292 31 01 – reception.  
+1 651 / 632-6793 (outside USA & Canada, collect call)

E-mail address : SDS-Info@fmc.com

#### Recommended use of the chemical and restrictions on use

Recommended use : Crop nutrition

Restrictions on use : Use as recommended by the label.  
For professional users only.

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### 2. HAZARDS IDENTIFICATION

#### GHS Classification

Not a hazardous substance or mixture.

Precautionary statements : **Prevention:**  
P280 Wear protective gloves, protective clothing.  
**Disposal:**  
P501 Dispose of contents/ container to an approved facility in accordance with local, regional, national and international regulations.

#### Other hazards which do not result in classification

None known.

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### 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Substance  
:

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## THYPAAN FERRO

Version 1.1      Revision Date: 18.12.2025      SDS Number: 50002035      Date of last issue: -  
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### Components

| Chemical name                                                                                        | CAS-No.    | Classification | MAC value<br>mg/m <sup>3</sup> /<br>TSEL value | Concentration (%<br>w/w) |
|------------------------------------------------------------------------------------------------------|------------|----------------|------------------------------------------------|--------------------------|
| Acetic acid, oxo-, sodium salt, reaction products with ethylenediamine and phenol, iron sodium salts | 84539-55-9 | Not classified | No data available                              | >= 90 - <= 100           |

No hazardous ingredients

### 4. FIRST AID MEASURES

- General advice : Move out of dangerous area.  
Show this safety data sheet to the doctor in attendance.  
Do not leave the victim unattended.
- If inhaled : Move to fresh air.  
If unconscious, place in recovery position and seek medical advice.  
If symptoms persist, call a physician.  
If experiencing any discomfort, immediately remove from exposure. Get medical attention if discomfort does not disappear.
- In case of skin contact : Take off all contaminated clothing immediately.  
Wash contaminated clothing before re-use.  
Wash off immediately with plenty of water for at least 15 minutes.  
Get medical attention if irritation develops and persists.
- In case of eye contact : Flush eyes with water as a precaution.  
Remove contact lenses.  
Protect unharmed eye.  
Keep eye wide open while rinsing.  
If eye irritation persists, consult a specialist.
- If swallowed : Do not induce vomiting without medical advice.  
Keep respiratory tract clear.  
Do not give milk or alcoholic beverages.  
Never give anything by mouth to an unconscious person.  
If symptoms persist, call a physician.
- Most important symptoms and effects, both acute and delayed : None known.
- Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing  
Avoid inhalation, ingestion and contact with skin and eyes.  
If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.

# SAFETY DATA SHEET



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|                |                              |                         |                                                          |
|----------------|------------------------------|-------------------------|----------------------------------------------------------|
| Version<br>1.1 | Revision Date:<br>18.12.2025 | SDS Number:<br>50002035 | Date of last issue: -<br>Date of first issue: 18.12.2025 |
|----------------|------------------------------|-------------------------|----------------------------------------------------------|

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### 5. FIREFIGHTING MEASURES

- |                                               |   |                                                                                                                                                            |
|-----------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media                  | : | Dry chemical, CO2, water spray or regular foam.<br>Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Unsuitable extinguishing media                | : | High volume water jet<br>Do not spread spilled material with high-pressure water streams.                                                                  |
| Hazardous combustion products                 | : | No hazardous combustion products are known                                                                                                                 |
| Specific extinguishing methods                | : | Standard procedure for chemical fires.<br>Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.          |
| Special protective equipment for firefighters | : | Firefighters should wear protective clothing and self-contained breathing apparatus.                                                                       |
- 

### 6. ACCIDENTAL RELEASE MEASURES

- |                                                                     |   |                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions, protective equipment and emergency procedures | : | Avoid dust formation.<br>Never return spills in original containers for re-use.<br>Mark the contaminated area with signs and prevent access to unauthorized personnel.<br>Only qualified personnel equipped with suitable protective equipment may intervene.<br>For disposal considerations see section 13. |
| Environmental precautions                                           | : | Prevent further leakage or spillage if safe to do so.<br>If the product contaminates rivers and lakes or drains inform respective authorities.                                                                                                                                                               |
| Methods and materials for containment and cleaning up               | : | Keep in suitable, closed containers for disposal.                                                                                                                                                                                                                                                            |
- 

### 7. HANDLING AND STORAGE

- |                                                 |   |                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advice on protection against fire and explosion | : | Provide appropriate exhaust ventilation at places where dust is formed.                                                                                                                                                                                                       |
| Advice on safe handling                         | : | Avoid formation of respirable particles.<br>Do not breathe vapours/dust.<br>For personal protection see section 8.<br>Smoking, eating and drinking should be prohibited in the application area.<br>Dispose of rinse water in accordance with local and national regulations. |
| Conditions for safe storage                     | : | Keep container tightly closed in a dry and well-ventilated place.                                                                                                                                                                                                             |
-

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|---------|----------------|-------------|---------------------------------|
| Version | Revision Date: | SDS Number: | Date of last issue: -           |
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Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : Keep in a dry place.  
No decomposition if stored and applied as directed.

### 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Components with workplace control parameters

##### Personal protective equipment

|                          |                                                                                                                                                                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Respiratory protection   | : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.                                                                                              |
| Filter type              | : Particulates type                                                                                                                                                                                                                                                            |
| Hand protection          |                                                                                                                                                                                                                                                                                |
| Material                 | : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.                                                                                                                                                                                    |
| Remarks                  | : The suitability for a specific workplace should be discussed with the producers of the protective gloves.                                                                                                                                                                    |
| Eye protection           | : Eye wash bottle with pure water<br>Tightly fitting safety goggles                                                                                                                                                                                                            |
| Skin and body protection | : Dust impervious protective suit<br>Choose body protection according to the amount and concentration of the dangerous substance at the work place.                                                                                                                            |
| Protective measures      | : Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions.<br>Ensure that eye flushing systems and safety showers are located close to the working place.<br>Wear suitable protective equipment. |
| Hygiene measures         | : Wash hands before breaks and at the end of workday.                                                                                                                                                                                                                          |

### 9. PHYSICAL AND CHEMICAL PROPERTIES

|                |                          |
|----------------|--------------------------|
| Physical state | : solid                  |
| Form           | : powder                 |
| Colour         | : black, red             |
| Odour          | : strong, characteristic |

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|----------------|------------------------------|-------------------------|----------------------------------------------------------|

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|                                                  |   |                                 |
|--------------------------------------------------|---|---------------------------------|
| Odour Threshold                                  | : | No data available               |
| pH                                               | : | 7.5 - 9.5<br>Concentration: 1 % |
| Melting point/freezing point                     | : | Decomposition at melting point. |
| Initial boiling point and boiling range          | : | Decomposition at boiling point. |
| Flash point                                      | : | No data available               |
| Upper explosion limit / Upper flammability limit | : | No data available               |
| Lower explosion limit / Lower flammability limit | : | No data available               |
| Vapour pressure                                  | : | < 0.000001 hPa (25 °C)          |
| Relative vapour density                          | : | No data available               |
| Relative density                                 | : | No data available               |
| Density                                          | : | 450 - 650 kg/m <sup>3</sup>     |
| Bulk density                                     | : | No data available               |
| Solubility(ies)                                  |   |                                 |
| Water solubility                                 | : | 60 - 120 g/l                    |
| Solubility in other solvents                     | : | No data available               |
| Partition coefficient: n-octanol/water           | : | log Pow: < 4.2 (23 °C)          |
| Auto-ignition temperature                        | : | 331 °C                          |
| Decomposition temperature                        | : | No data available               |
| Viscosity                                        |   |                                 |
| Viscosity, dynamic                               | : | No data available               |
| Viscosity, kinematic                             | : | No data available               |
| Explosive properties                             | : | Not explosive                   |
| Oxidizing properties                             | : | Non-oxidizing                   |
| Particle size                                    | : | No data available               |

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---

**10. STABILITY AND REACTIVITY**

|                                    |   |                                                                                                |
|------------------------------------|---|------------------------------------------------------------------------------------------------|
| Reactivity                         | : | No decomposition if stored and applied as directed.                                            |
| Chemical stability                 | : | No decomposition if stored and applied as directed.                                            |
| Possibility of hazardous reactions | : | No decomposition if stored and applied as directed.<br>Dust may form explosive mixture in air. |
| Conditions to avoid                | : | Avoid dust formation.<br>Avoid extreme temperatures                                            |
| Incompatible materials             | : | Avoid strong acids, bases, and oxidizers                                                       |
| Hazardous decomposition products   | : | No hazardous decomposition products are known.                                                 |

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**11. TOXICOLOGICAL INFORMATION****Acute toxicity**

Based on available data, the classification criteria are not met.

**Product:**

|                           |   |                                                                                                                 |
|---------------------------|---|-----------------------------------------------------------------------------------------------------------------|
| Acute oral toxicity       | : | LD50 Oral (Rat): > 2,000 mg/kg<br>Method: OECD Test Guideline 401                                               |
| Acute inhalation toxicity | : | LC50 (Rat): > 4,200 mg/l<br>Exposure time: 4 h<br>Test atmosphere: dust/mist<br>Method: OECD Test Guideline 403 |
| Acute dermal toxicity     | : | LD50 Dermal (Rat): > 2,000 mg/kg<br>Method: OECD Test Guideline 402                                             |

**Skin corrosion/irritation**

Based on available data, the classification criteria are not met.

**Product:**

|            |   |                            |
|------------|---|----------------------------|
| Species    | : | Rabbit                     |
| Assessment | : | Not classified as irritant |
| Method     | : | OECD Test Guideline 404    |
| Result     | : | No skin irritation         |

**Serious eye damage/eye irritation**

Based on available data, the classification criteria are not met.

**Product:**

|            |   |                            |
|------------|---|----------------------------|
| Species    | : | Rabbit                     |
| Result     | : | No eye irritation          |
| Assessment | : | Not classified as irritant |
| Method     | : | OECD Test Guideline 405    |

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|---------|----------------|-------------|---------------------------------|
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---

**Respiratory or skin sensitisation****Skin sensitisation**

Based on available data, the classification criteria are not met.

**Respiratory sensitisation**

Based on available data, the classification criteria are not met.

**Product:**

|            |   |                                                    |
|------------|---|----------------------------------------------------|
| Test Type  | : | Local lymph node assay (LLNA)                      |
| Species    | : | Rat                                                |
| Assessment | : | Not classified as irritant                         |
| Method     | : | OECD Test Guideline 429                            |
| Result     | : | Did not cause sensitisation on laboratory animals. |

**Components:****Acetic acid, oxo-, sodium salt, reaction products with ethylenediamine and phenol, iron sodium salts:**

|           |   |                         |
|-----------|---|-------------------------|
| Test Type | : | Maximisation Test       |
| Species   | : | Guinea pig              |
| Method    | : | OECD Test Guideline 406 |
| Result    | : | equivocal               |

|           |   |                         |
|-----------|---|-------------------------|
| Test Type | : | Local lymph node test   |
| Method    | : | OECD Test Guideline 429 |
| Result    | : | equivocal               |

**Germ cell mutagenicity**

Based on available data, the classification criteria are not met.

**Carcinogenicity**

Based on available data, the classification criteria are not met.

**Reproductive toxicity**

Based on available data, the classification criteria are not met.

**STOT - single exposure**

Based on available data, the classification criteria are not met.

**STOT - repeated exposure**

Based on available data, the classification criteria are not met.

**Aspiration toxicity**

Based on available data, the classification criteria are not met.

**Further information****Product:**

|         |   |                   |
|---------|---|-------------------|
| Remarks | : | No data available |
|---------|---|-------------------|

# SAFETY DATA SHEET



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### 12. ECOLOGICAL INFORMATION

**Ecotoxicity**

No data available

**Persistence and degradability**

No data available

**Bioaccumulative potential**

No data available

**Mobility in soil**

No data available

**Other adverse effects****Product:**

Additional ecological information : No data available

### 13. DISPOSAL CONSIDERATIONS

**Disposal methods**

Waste from residues : Do not dispose of waste into sewer.  
Do not contaminate ponds, waterways or ditches with chemical or used container.  
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.  
Dispose of as unused product.  
Do not re-use empty containers.

### 14. TRANSPORT INFORMATION

**ADR**

Not regulated as a dangerous good

**UNRTDG**

Not regulated as a dangerous good

**IATA-DGR**

Not regulated as a dangerous good

**IMDG-Code**

Not regulated as a dangerous good

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable for product as supplied.

**Special precautions for user**

Remarks : Not classified as dangerous in the meaning of transport regulations.

|         |                |             |                                 |
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## 15. REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**The components of this product are reported in the following inventories:**

|       |                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TCSI  | : On the inventory, or in compliance with the inventory                                                                                                                                                 |
| TSCA  | : Product contains substance(s) not listed on TSCA inventory.                                                                                                                                           |
| AIIC  | : Not in compliance with the inventory                                                                                                                                                                  |
| DSL   | : This product contains the following components that are not on the Canadian DSL nor NDSL.<br><br>Acetic acid, oxo-, sodium salt, reaction products with ethylenediamine and phenol, iron sodium salts |
| ENCS  | : Not in compliance with the inventory                                                                                                                                                                  |
| ISHL  | : Not in compliance with the inventory                                                                                                                                                                  |
| KECI  | : Not in compliance with the inventory                                                                                                                                                                  |
| PICCS | : Not in compliance with the inventory                                                                                                                                                                  |
| IECSC | : Not in compliance with the inventory                                                                                                                                                                  |
| NZIoC | : Not in compliance with the inventory                                                                                                                                                                  |
| TECI  | : Not in compliance with the inventory                                                                                                                                                                  |

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## 16. OTHER INFORMATION

### Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Con-

# SAFETY DATA SHEET



## THYPAAN FERRO

|         |                |             |                                 |
|---------|----------------|-------------|---------------------------------|
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| 1.1     | 18.12.2025     | 50002035    | Date of first issue: 18.12.2025 |

---

centration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

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